

equipped with the smart wake-up feature that enables it to wake only at defined intervals and scan the channels to receive a valid packet. It allows the designers to incorporate timing-critical elements so as to facilitate shorter deployment cycles. Such transceivers are designed for high-performance cost-effective wireless systems operating at very low-power and low-voltage conditions. These have integrated filters and so do not require any of the costly external filters.

Operational amplifiers (op-amps)

Op-amp is another important component that helps in wireless communication. It captures the sensor information, amplifies it and then passes it on.

The latest range of op-amps offer very low input offset voltage and almost zero drift. Thus, being high-precision op-amps, these allow the engineers to design a product that would process the data without much wait. This family of op-amps is virtually unaffected by any temperature change and can tolerate extreme temperatures in the range of -40°C to +125°C.

New-generation system on chip (SoC)

From BLE to the other standards, such as IEEE 802.15.4, RF4CE and ZigBee, the SoCs are compatible with many applications based on these standards. There are highly integrated SoC solutions with tightly coupled physical layer of the OSI model (PHY) and media access control (MAC) to minimise latency and to avoid complex software partitioning across multiple devices. The best thing is the availability of better software suites to support the design of these SoCs.

Safe and secure communication parameters

The increasing use of wireless communication brings security into question. Then there are issues like noise, interference, energy efficiency, network coverage, reliability and latency. Purkayastha says, "Several integrated circuits with interference management

TABLE II Design and Development Tools		
Name	Type	Manufacturer
Code Composer Studio (CCS) Integrated Development Environment (IDE) for Multicore Processors	SW Development Tools, IDEs, Compilers	Texas Instruments
WLAN Measurement Suite	Test and Measurement tool Support for IEEE 802.11a/b/g/n/p/ac standards and up to 4x4 MIMO measurements	National Instruments
DriveExecutive software	Provides a simplified interface for programming, maintenance and troubleshooting of your PowerFlex AC and DC drives	Allen Bradley
Visual System Simulator	Enables engineers to design the right system architecture and formulate suitable specifications for each of the underlying components.	National Instruments
Advanced Design System	EDA software for RF, microwave, and high speed wireless communication with WiMAX, LTE, multi-gigabit per second data links, radar and satellite applications	Keysight Technologies
SimpleLink Bluetooth CC2564 Module Evaluation Board	Evaluation Modules and Boards	Texas Instruments

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and noise filtering capabilities are used in communication devices. Interference is a major issue in crowded areas, such as shopping malls, airports, busy industrial areas, etc, where MIMO with multiple antennae for transmission and reception are used for interference cancellation." Satish Mohanram, technical marketing manager, National Instruments, says, "High-end wireless technologies like 5G require high amount of bandwidth to transfer large amount of data without any latency. MIMO provides those devices a space diversity for a deterministic transfer for data over wireless medium."

Companies like National Instruments are working on the prototyping tools for Massive MIMO to help design high-end wireless technology based devices from the scratch. Massive MIMO enables MIMO diversity and transmission without any loss in signal, enables the use of low-cost and low-power components, simplifies the media access control (MAC) layer and also provides robustness to intentional jamming and interference.

Test initiatives and solutions

Cost-effective ways have come up recently to test wireless communication devices at a high speed using shorter and flexible measurement steps. The tools not only test the existing but also the new formats of radio, including WiMAX, Bluetooth, HSPA+ and LTE. Then there are test sets to reduce the capital investment required in order to optimise the architecture for the non-signalling test. These sets help plan the test and troubleshooting graphically, which simplifies the testing process and reduces long-term costs related to the advancements in test modes.

From latency to high-power usage, the new range of embedded components is helping the designers to overcome all the issues that would decrease the popularity of the wireless devices designed by them. It also opens up a lot of opportunities before the designers to choose the best components at a very low price, and the changing technologies are helping further. ●

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